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Evaluation of Tailor-Made Generative AI Solutions for Efficient Test Vehicle Allocation at Volvo Cars

- A Case Study on Feasibility, Effort, and Implementation Practices

Ayeman Ulfat

UNIVERSITY WEST

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Date:	June 5, 2026
Author:	Ayeman Ulfat
Examiner:	Morgan Nilsen
Advisor:	Fredrik Ahrenberg, Volvo Cars
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Summary

The thesis addresses Volvo Cars' interest in developing systematic and reusable frameworks for evaluating existing business processes. The aim was to better understand where agentic AI can deliver real value, where simpler forms of automation would suffice, and where human-led processes remain the best option.

The main goal was to create a practical framework to evaluate feasibility and business value, gauge the actual time and effort needed to reach satisfactory domain-specific performance, and find effective methods and best practices for future AI projects across the organization.

To this end, I have developed the Volvo Hybrid AI Adoption and Readiness Framework (V-HAARF). This multi-level model integrates strategic considerations, process-level readiness assessment, and detailed task analysis. It provides the organisation with a consistent and informed basis for deciding how to approach AI adoption in different areas.

In parallel, I designed and applied a focused three-layer evaluation framework specifically for assessing agentic AI solutions already under implementation. I tested this framework through a detailed case study of the CTPBot project, an agentic AI tool introduced to support the Compatible Test Plan workflow for test vehicle allocation. The case study served as real-world validation for the post-implementation framework and yielded concrete insights into actual effort, challenges, and outcomes.

Collectively, these contributions provide Volvo with a pragmatic basis: a wide decision-making framework for future AI initiatives and a tried-and-tested evaluation method based on hands-on experience with the CTPBot project. I believe this work can help the company in pursuing AI adoption in a more strategic, efficient, and responsible way.

Preface

This master's thesis project has been completed in the framework of the Master's Programme in Artificial Intelligence and Automation at University West in Sweden. I began this thesis project in January 2026 as a thesis worker at Volvo Cars in Torslanda. The opportunity came at just the right moment. I had long wanted to explore how generative and agentic AI could be introduced into a real, complex industrial process, and the collaboration between Volvo Cars and its external partner offered exactly that setting.

From the first week, I was given full access to the ongoing Automation Verification work. I joined the daily evaluation of the CTPBot, tested every major operation in the live CTP environment, logged issues as they appeared, and tracked my own effort hour by hour. The work was demanding, but it was also deeply rewarding. I quickly realised that success in such projects depends as much on careful validation and knowledge capture as on the technology itself.

I could not have completed this thesis without the generous support of many people. My Volvo supervisor guided me through the complexity of the CTP workflow and gave me honest feedback whenever my findings needed sharpening. My university supervisor encouraged me to keep the academic perspective clear while staying grounded in the practical realities of the project. The development team from the external partner responded quickly to every issue I reported and openly discussed both successes and setbacks. I am also grateful to my thesis partner, who shared the perspectives on our joint analysis.

This experience has taught me that introducing agentic AI into established engineering workflows is never a simple technology swap. It is a process of learning, adjustment, and continuous refinement. I hope the findings and the hybrid evaluation framework presented in this report will prove useful to Volvo Cars and to others who face similar challenges in the years ahead.

Affirmation

This master degree report, *Evaluation of Tailor-Made Generative AI Solutions for Efficient Test Vehicle Allocation at Volvo Car- A Case Study on Feasibility, Effort, and Implementation Practices,* was written as part of the master degree work needed to obtain a Master of Science with specialization in Artificial Intelligence and Automation degree at University West. All material in this report, that is not my own, is clearly identified and used in an appropriate and correct way. The main part of the work included in this degree project has not previously been published or used for obtaining another degree.

Ayeman Ulfat

2026-05-27

Signature by the author | Date

Ayeman Ulfat

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Main Chapters

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List of abbreviations

Abbreviation	**Full Form / Explanation**
AI	Artificial Intelligence
Agentic AI	AI systems that can perceive, reason, plan, and act autonomously
CTP	Compatible Test Plan
CTPBot	The agentic AI solution (chatbot) developed to support the CTP workflow
SLO	Service Level Objective – measurable performance and governance targets
API	Application Programming Interface
LLM	Large Language Model
RAI	Responsible AI
CSFs	Critical Success Factors
DSR	Design Science Research
LARA	LLM Agent Readiness Assessment
RPA	Robotic Process Automation
T-IPO	Task Input-Process-Output
V-HAARF	Volvo Hybrid AI Adoption & Readiness Framework

1 Introduction

The automotive industry is undergoing a fundamental transformation. Digital technologies, particularly artificial intelligence, are reshaping the way engineering and operational processes are designed and executed.

This transformation goes hand in hand with strong ambitions in electrification, software-defined vehicles, and sustainable mobility for Volvo Cars. Yet the company shows interest on how to adopt AI intelligently across its many complex processes. Not every process gains the same benefit from advanced agentic AI. In some cases, traditional automation proves more suitable and cost-effective. In others, human-led approaches remain the better choice. Making these distinctions consistently and thoughtfully matters greatly. It allows the company to get real value, without wasting effort. This thesis tackles that challenge. I developed a general framework to support more informed and structured decisions about AI adoption at Volvo Cars.

1.1 The Role of Digitalisation and Automation in Automotive Development

Automotive development today involves complex design, verification, manufacturing support, and conformance processes that must address exponentially growing data volumes, complex interdependency, and constantly changing requirements, all within the constraints of strict safety, quality, and regulatory standards [1]. The explosion of software-defined, feature-rich vehicles makes the need to reduce development cycles and improve resource efficiency more urgent than ever.

In this context, digital support has become not only useful but strategically vital. Traditional rule-based automation has served well for repetitive tasks for many years. However, many processes in automotive engineering involve variable inputs, contextual judgment, and dynamic decision making. Deciding the right level of digital support, whether simple automation, advanced agentic systems, or continued human oversight, carries real consequences for development speed, cost, risk, and long-term maintainability [2].

Getting this balance wrong can waste resources or cause missed opportunities. For companies such as Volvo Cars, a structured and thoughtful approach to evaluating which processes suit different digital solutions has therefore become increasingly important.

1.2 The Emergence of Generative and Agentic AI in Engineering Workflows

Recent advances in generative AI and agentic systems have introduced powerful new capabilities that extend well beyond traditional automation. While conventional systems operate according to predefined rules and fixed logic, agentic AI can reason through ambiguous situations, interpret varied and unstructured inputs, adapt its behaviour based on feedback, and pursue higher-level goals with a degree of autonomy [3]. These characteristics make such systems particularly interesting for complex engineering environments where rigid automation often falls short.

In industrial and engineering contexts, agentic AI shows strong potential for supporting tasks such as resource optimisation, scenario analysis, and contextual decision support. However, deploying these technologies successfully requires more than technical sophistication. Factors such as organisational readiness, the realistic effort needed to build sufficient domain knowledge, and the establishment of clear governance practices play a decisive role in determining outcomes [8]. Understanding how and when to apply agentic AI, as opposed to simpler automation, remains a critical challenge for many organisations.

1.3 Problem Description at Volvo Cars

Volvo Cars' operational and engineering processes involve a mix of manual work and varying levels of automation. At the same time, interest in agentic AI solutions continues to grow. A standardised method aims to ensure that important factors are not considered in an informal or ad-hoc manner. Figure 1 includes the nature of data flows, whether tasks follow fixed rules or demand interpretation and reasoning, the level of process debt, compliance requirements, and the real effort needed to build sufficient domain knowledge.

This approach can result in suboptimal decisions. Teams might be building complex AI when a simpler solution would do, or they might be missing opportunities where agentic AI could improve things.

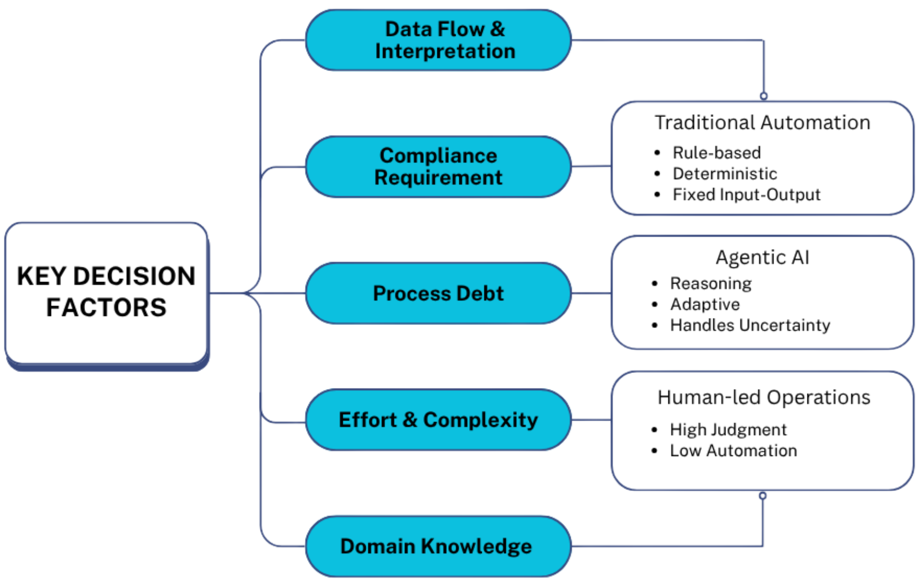

Figure 1 Decision Challenge in AI Adoption at the company

Figure 1 summarises the main decision factors that Volvo teams face when evaluating digital solutions. It highlights that important considerations, such as data flow characteristics, compliance requirements, process debt, effort, complexity, and domain knowledge, are often assessed informally today.

1.4 Purpose and Aims of the Thesis

The main goal of this thesis was to develop a practical and reusable framework to help Volvo Cars to evaluate current business processes and support better decision-making regarding the most suitable type of digital support, such as agentic AI, traditional automation, or mostly human-led operations.

The specific aims were:

- To create the Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF) as a multi-level decision-support tool applicable across the company.
- To define clear methods for assessing feasibility, business value, implementation complexity, and required domain knowledge.
- To identify best practices and produce supporting tools that facilitate consistent evaluation of future AI initiatives.
- To demonstrate the framework's logic and structure through analysis grounded in real industrial conditions.

1.5 Investigative Questions

To achieve this aim, the thesis is guided by three focused investigative questions that directly address the core dimensions of the project:

- To what extent did the agentic generative AI solution bring real feasibility and business value to the workflow? I examined this through practical measures such as how accurately it assigned tasks, how much manual work it cut, whether vehicle utilisation improved, and how effectively the team could adopt the new approach.
- How much time, resources, and effort were required to reach a satisfactory level of domain-specific performance? This included everything from initial development and integration through validation, refinement, and the final knowledge transfer phases.
- What methods, best practices, and working mechanisms emerged from the project that supported successful implementation and knowledge sharing? I looked at how we could turn the useful elements, such as our back-end storage solutions and feedback loops, into clear, reusable frameworks for other AI initiatives at Volvo Cars.

These questions guided my decisions from the beginning and kept the work grounded in what mattered for both the immediate project and future efforts inside the company.

1.6 Scope and Limitations

The scope of this thesis is to develop a general evaluation framework and not to deploy any specific AI system. Although my work has been inspired by insights from ongoing AI projects at Volvo, the final deliverable is a reusable reference model for process evaluation and decision making. My design of the framework has been done with adaptability in mind across different areas of the company.

The study had clear limitations. The timeframe of the thesis restricted the depth of validation I could achieve. In addition, I developed the framework mainly through literature synthesis supported by a single contextual case study. Broader testing across multiple processes and domains will therefore be necessary in future work.

1.7 Thesis Structure and Overview

The thesis is divided into ten chapters. Each chapter is based on the previous one and moves from context and theory to the development and application of the framework, to analysis and conclusions.

In chapter 1 I describe the wider role of digitalisation in car development and the rise of agentic AI and set the stage for the study. I lay out the specific challenges Volvo Cars is facing and present the purpose and aims for the study. Lastly, I introduce the investigative questions that have guided my work throughout the study.

Chapter 2 reviews the relevant literature. This chapter discusses agentic AI, process readiness frameworks, task level evaluation approaches and existing decision models for AI vs automation. It also highlights important gaps and how this thesis aims to fill them.

In Chapter 3, I present the research methodology. I describe the design science-inspired approach used to develop the frameworks and explain the qualitative and quantitative techniques applied during analysis and validation.

The main contribution of the thesis is presented in Chapter 4: the Volvo Hybrid AI Adoption and Readiness Framework (V-HAARF). I describe its core concept, key definitions, three-layer architecture and how it can be applied in different parts of the company.

Chapter 5 is about the practical use of V-HAARF. It describes the step-by-step procedure to use the framework, best practices, adaptation to different domains and the value it can bring to Volvo Cars.

In Chapter 6 I introduce the second framework developed in this thesis, a focused post-implementation evaluation framework for agentic AI systems. I explain its rationale, theoretical foundations, three-layer structure, scoring method, practical guidelines, and the supporting tools and templates I created.

Chapter 7 reports the results of applying this post-implementation framework to a real AI initiative at Volvo Cars. It presents the empirical data collected, shows how the framework mapped onto the actual workflow, and discusses the scoring outcomes and insights gained from the supporting tools.

Chapter 8 analyses and discusses both frameworks in depth. I reflect on their feasibility and business value, examine the effort required to build domain-specific knowledge, review the methods and best practices that emerged, and consider how the frameworks relate to existing literature.

In Chapter 9 the key findings are summarised, contributions to Volvo Cars and the academic knowledge are highlighted, limitations of the study are acknowledged and practical recommendations are provided.

Finally, Chapter 10 discusses directions for future work, including further empirical validation, tool development and a wider organisational uptake of the frameworks.

2 Related Work and Theoretical Background

Adopting agentic AI in large industrial organisations involves far more than technical implementation. It requires a clear and structured understanding of when such systems truly add value over traditional automation, how much real effort they demand, and which practices best support successful deployment.

This chapter reviews the relevant literature in several areas: agentic AI applications, process assessment frameworks, task-level evaluation methods, and existing decision models for choosing between AI, automation, and human-led approaches.

2.1 Foundations of Agentic AI and Its Role in Industrial Processes

Agentic AI is AI that can pursue goals on its own. They reason through uncertainty, use external tools, and adapt their behaviour based on feedback [3], [8]. Unlike traditional rule-based automation, these systems show a much higher degree of autonomy. They can manage non-deterministic tasks that involve interpretation and contextual judgment.

In industrial environments, such capabilities have demonstrated real potential. Applications include process optimisation, resource allocation, and complex decision support [1].

Still, successful deployment depends heavily on the organisational context. Recent studies suggest that many AI initiatives fall short not because of technical shortcomings, but due to poor assessment of process readiness and overly optimistic assumptions about the effort and domain knowledge required [11], [12].

2.2 Process Assessment and Readiness Frameworks

Several frameworks have been proposed to evaluate organisational or process readiness for AI adoption. Ragsdale's Framework for Autonomous Organisations (RFAO) introduces a phased evolution model and multi-dimensional structures (Actor, Decision, Story, Structure, and Control Models) to guide organisations toward higher levels of autonomy [16]. This perspective is useful in considering the long-term trajectory of AI integration.

Complementing this, Schmidt et al. (2026) propose a process-oriented readiness assessment that analyses five key perspectives, activities, decisions, data operations, control flow, and resource management, while accounting for "process debt" (the gap between documented and actual practices) [17]. Their work emphasises that practical readiness is often significantly lower than theoretical potential.

2.3 Task-Level AI Evaluation Approaches

At the task level, Li and Ye (2026) developed two practical tools for single-task evaluation, T-IPO model (Task Input-Process-Output) and LARA matrix (LLM Agent Readiness Assessment) [17], to help make better-informed decisions on whether a particular task should be best performed by agentic AI, human-AI collaboration, or existing automation. The LARA matrix stands out with its inclusion of a weighted compliance sensitivity factor, adding value to regulated industrial settings.

2.4 Decision Models for AI versus Automation

The literature increasingly emphasises the importance of having clear decision criteria when choosing between agentic AI and simpler forms of automation. Key considerations typically include how deterministic the data flows are, whether the task requires interpretation and judgment, compliance demands, and the overall cost-benefit balance [17], [18].

Frameworks that effectively combine strategic, process, and task-level perspectives are still relatively uncommon. This gap gives a valuable opportunity for more holistic approaches, which is exactly what I set out to do in this thesis.

2.5 Gaps in Current Research and Relevance to Volvo's Needs

Existing frameworks offer useful insights, but few combine strategic considerations, process-level readiness assessment, and task-level decision tools into one practical model suitable for large industrial organisations. Guidance on estimating the real effort and domain knowledge required across different processes is still limited.

This thesis addresses these gaps. Figure 2 shows the proposed Volvo Hybrid AI Adoption and Readiness Framework (V-HAARF), which integrates and extends key ideas from the reviewed literature. The framework is intended to support more consistent and informed decisions on AI adoption at Volvo Cars.

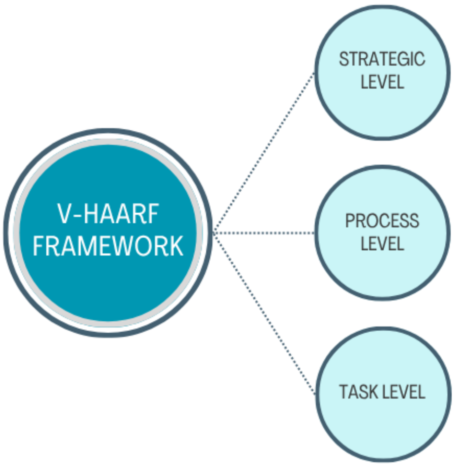

Figure 2 Conceptual integration diagram

Figure 2 presents the overall conceptual structure of V-HAARF. It brings together three complementary levels of analysis for a holistic assessment of any business process.

3 Method

This chapter describes the methodological approach used to develop the Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF). Given the scope of creating a general, reusable framework applicable to any business process at Volvo Cars, the research followed a design-oriented and iterative process. The aim was not only to propose a theoretical model but to create a practical tool that can support real decision-making regarding agentic AI versus traditional automation across different domains.

3.1 Research Approach

This study adopted a mixed-methods, iterative Design Science Research (DSR) approach [19], [20]. DSR is particularly suitable when the goal is to create and evaluate an artefact (in this case, a framework) that solves a practical organisational problem while contributing to scientific knowledge. The research employed a combination of synthesis of the literature, conceptual modelling and practical validation techniques to ensure the framework is both theoretically grounded and applicable in an industrial context. The development was based on an iterative cycle comprising five main steps:

- Process mapping and task decomposition using the T-IPO model.
- Multi-stakeholder-informed scoring based on process perspectives and readiness criteria.
- Quantitative scoring combined with qualitative validation.
- Conceptual pilot application on selected high-readiness scenarios.
- Continuous refinement and recalibration of the framework.

This structured yet flexible approach enabled the framework to tackle the main aspects of the expanded scope: feasibility assessment and business value, complexity and effort estimation, and identification of suitable methods and best practices.

3.2 Literature Synthesis and Theoretical Foundation

To start this research, I performed a thorough review of the existing literature on agentic AI, process readiness assessment, task-level evaluation, and organisational approaches to increasing autonomy. The key sources included frameworks for strategic AI adoption [16], process-oriented readiness models [18], and more granular task assessment methods [17].

These works supplied the main theoretical building blocks. I synthesised and extended them to form the foundation of the Volvo Hybrid AI Adoption and Readiness Framework (V-HAARF).

3.3 Framework Development Process

Figure 3 describes the heart of the work, which consisted of iterative design activities. I started by conceptually mapping and decomposing processes into individual tasks using the T-IPO approach. I then evaluated readiness at three levels: strategic, process, and task.

Scoring mechanisms emerged gradually as I combined multi-perspective analysis with weighted evaluation matrices. I have also refined the decision criteria along the way with qualitative insights from real industrial contexts, especially when agentic AI has clear advantages over traditional automation.

Figure 3 Core Architecture of the Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF)

Figure 3 illustrates the core architecture of V-HAARF. At all levels of decision-making, from strategy to task, three related layers inform it.

3.4 Qualitative Analysis

I conducted qualitative analysis that shed light on the contextual and organizational drivers of AI adoption. I used Template Analysis [18] to find common themes across the literature that related to process debt, sensitivity to compliance, and the need for effective human-AI collaboration.

I also used multi-perspective examination techniques as described by Schmidt et al. to gain insights into the different process characteristics impacting suitability for agentic AI. In doing so, I had in mind stakeholder considerations and emerging best practices. I tried to keep the framework grounded in the practical challenges we find in industrial settings.

3.5 Quantitative Analysis

In this work, I have used quantitative methods to make the framework more quantifiable and comparable between processes. I have developed a weighted scoring system by combining parts of the LARA matrix with different readiness scales.

I had numerical scores against the key dimensions of compliance sensitivity, effort estimation, and overall readiness levels. This gave me definitive AI-worthiness scores that can help in prioritisation. The quantitative aspect complements the qualitative insights. "It converts observations into metrics that teams can use to determine time, resources, and effort to put in.

3.6 Validation and Evaluation Techniques

I validated the framework by several approaches, namely logical consistency checks, cross comparison with well-established models from the literature, and retrospective application to real industrial conditions I had experienced. I used a four-point validation method to evaluate four key aspects of completeness, practicality, adaptability, and alignment with organisational needs.

For the effort estimation, I used some existing literature on the complexity of AI implementation. Then I adjusted the estimates according to typical patterns observed in industrial projects. This was useful to make the predictions more realistic and grounded.

While the V-HAARF framework remains conceptual, the post-implementation evaluation framework was applied to the CTPBot project. The results of this empirical application are presented in Chapter 7.

4 Design and Architecture of the Proposed Framework

This chapter presents the main contribution of the thesis: the Volvo Hybrid AI Adoption and Readiness Framework, or V-HAARF. I developed this framework to support Volvo Cars' systematic scope and reusable method. It helps evaluate any existing business process and reach well-informed decisions about the most suitable approach, whether agentic AI, traditional automation, or primarily human-led operations.

The framework directly addresses the scope of this work. It provides practical guidance on how to determine what is possible and worthwhile for the business, estimate the complexity and effort needed, and identify what works and best practices for future AI efforts.

4.1 Main Idea of the Framework

The Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF) is a practical, multi-level decision-support tool designed to help Volvo Cars evaluate any business process and determine whether it should adopt agentic AI (LLM-based intelligent systems), simpler rule-based automation (RPA), or remain primarily human-led [16], [18].

The framework operates across three integrated levels of analysis, allowing both high-level strategic consideration and detailed operational decision-making [17]:

- **Strategic Level:** Guides the overall organisational evolution toward higher autonomy using a phased roadmap. It helps leadership understand the long-term journey and maturity required to adopt AI [16] successfully.
- **Process Level:** The structural maturity of a complete process is assessed from important facets, such as activities, decisions, data operations, control flow, and resource management, considering process debt (differences between documented and actual practices) [18].
- **Task Level:** Detailed evaluation of individual tasks in a process to find the most appropriate level of intelligence – from fully agentic AI to simple automation or human execution [17].

By bringing these three levels together, V-HAARF helps Volvo Cars more systematically identify where agentic AI can deliver genuine value. At the same time, it supports the company in avoiding investments in processes that are not yet mature enough or do not need advanced intelligence.

The framework places strong emphasis on realistic evaluation. It examines feasibility, expected business value, implementation complexity in terms of time and effort, and the actual domain knowledge required for successful deployment [3]. I've designed it to be flexible across departments and processes so it can be used as a reusable reference for future AI initiatives across the company.

This hybrid approach fills an important gap in current practice. It moves away from particular decision-making toward a more consistent and structured methodology for AI adoption at Volvo Cars.

4.2 Key Definitions

To ensure clarity and consistency, I defined several core concepts that underpin the framework. These definitions draw on recent literature but have been adapted to suit Volvo Cars' practical needs when evaluating processes for agentic AI adoption.

AI Worthiness

AI Worthiness describes the overall suitability of applying agentic AI to a particular process or task. It emerges from a balanced evaluation of feasibility, expected business value, implementation complexity in terms of time and effort, the domain knowledge that must be developed, and governance readiness. This helps teams understand where agentic AI can really add value, and where automation or human-driven approaches might be better [3], [8].

Process Debt

Process Debt refers to the gap between formally documented procedures and how work is carried out in practice. It includes inconsistencies in data flows, undocumented variations, unclear decision boundaries, and resource mismatches. High process debt significantly lowers a process's readiness for agentic AI and typically needs to be addressed before successful implementation becomes realistic [18].

Task Atomization

Task atomization is the process of breaking down larger activities into smaller, well-defined tasks, following the T-IPO model (Task – Input – Process – Output). This decomposition allows a more accurate assessment of each task. Such an assessment can help determine whether the work is rule-based and therefore suitable for traditional automation, or whether it requires interpretation, reasoning and adaptability, suggesting the need for agentic AI [17].

Autonomization

Autonomization refers to the gradual organizational transition from processes that require a high degree of human coordination to those where AI is taking a greater role in execution. Importantly, human strategic oversight remains in place. This concept captures the longer-term evolution toward higher levels of intelligent automation across different areas of the company [16].

Additional Key Concepts (Agentic AI vs. Traditional Automation)

Agentic AI is a system that can think, deal with ambiguity, use tools, and pursue goals to a significant degree of independence. Traditional automation, such as robotic process automation (RPA), is built on a static, rule-based logic with predictable input and output. The framework supports clear decisions by analysing task characteristics and organisational context to determine which approach fits best [3], [17].

Domain-Specific Knowledge

This refers to the specialised understanding of Volvo's processes, data structures, compliance requirements, and operational nuances needed for effective AI performance. The framework explicitly considers the effort required to develop this knowledge as a core component of AI adoption decisions.

Readiness Assessment

Figure 4 highlights a structured evaluation that combines potential readiness (theoretical suitability) and practical readiness (accounting for current constraints such as process debt and governance maturity) to support go/no-go decisions [17], [18].

These definitions form the conceptual foundation of V-HAARF and ensure that the framework remains focused on Volvo's strategic goal: making consistent, evidence-based decisions about intelligent automation across diverse processes in the company.

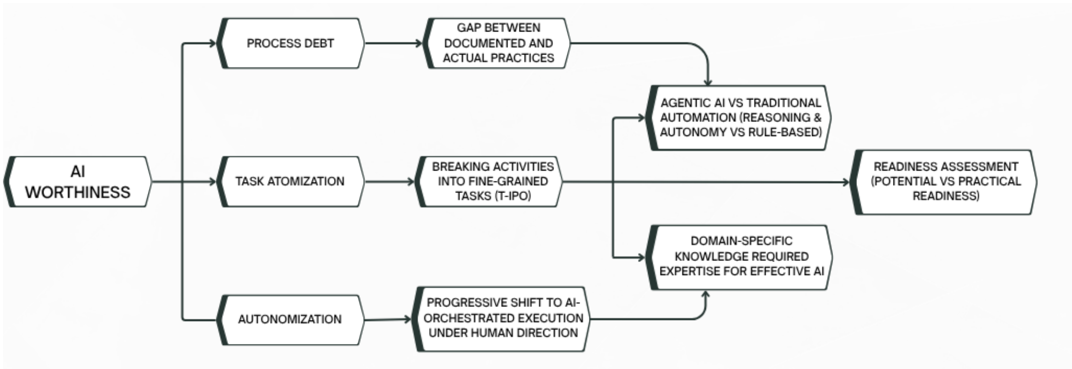

Figure 4 Core Concepts and their Relationships in the V-HAARF Framework (Developed by the author based on Ragsdale [16], Schmidt et al. [18], and Li & Ye [17])

The diagram in Figure 4 presents the centrality of AI Worthiness and the connections between key concepts such as Process Debt, Task Atomization, and Autonomization.

4.3 Architecture of the Framework

The Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF) is based on a three-layered model. The layered structure was deliberately chosen to accommodate by linking high-level strategic thinking with practical, operational decision-making. Each layer deals with a different level of analysis; however, they are closely integrated. Together, they provide a comprehensive evaluation mechanism that helps Volvo determine the most suitable form of digital support for any business process, whether agentic AI or traditional automation.

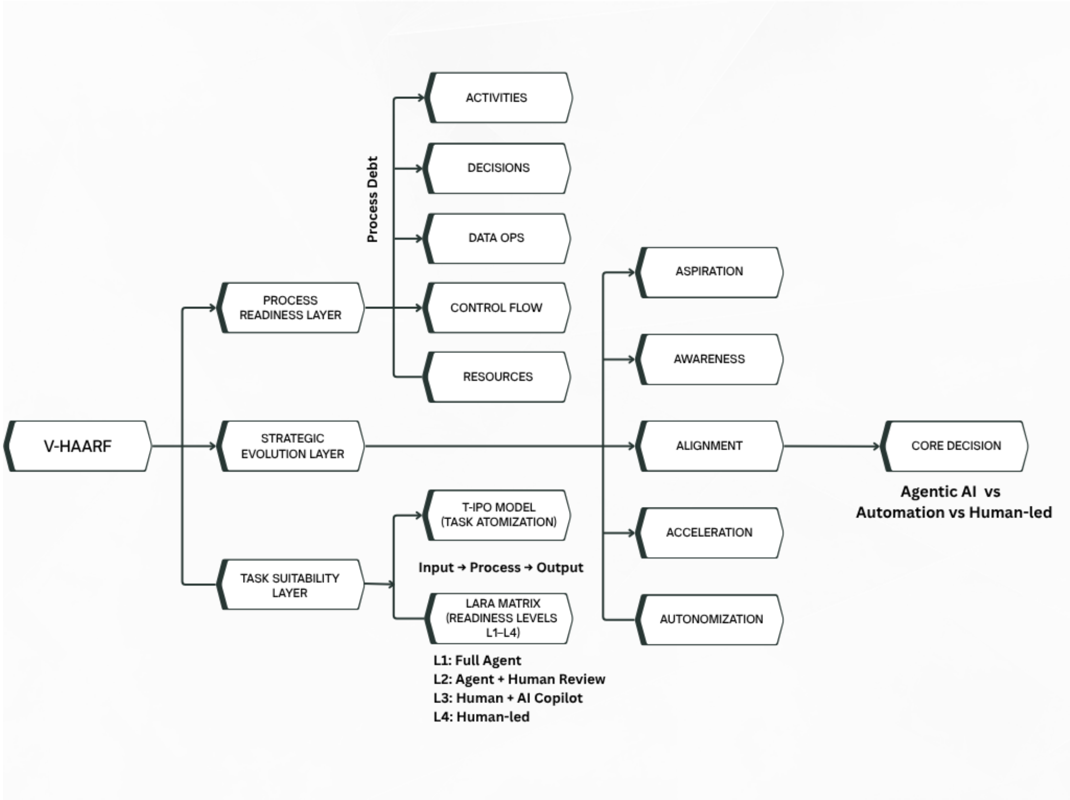

Figure 5 Overview of the Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF). The framework integrates three levels: Strategic Evolution (adapted from Ragsdale, 2025), Process Readiness (adapted from Schmidt et al., 2026), and Task Suitability (adapted from Li & Ye, 2026)

Figure 5 illustrates an outline of the entire V-HAARF framework and the support provided by the three layers for a holistic assessment.

Layer 1 – Strategic Evolution Layer

This uppermost layer establishes the long-term organisational perspective. Drawing upon Ragsdale's evolution model [16], it outlines five progressive phases that organisations typically move through when increasing their use of intelligent systems:

1. **Aspiration** - Leadership recognises the strategic potential of AI and commits to exploring its possibilities.
2. **Awareness** - The organisation builds visibility into current processes, data maturity, capabilities, and existing gaps.
3. **Alignment** - Critical foundational work takes place, including unifying data standards, updating policies, and establishing governance structures.
4. **Acceleration** - AI solutions are piloted and scaled in a governed manner.
5. **Autonomization** - AI systems begin orchestrating complex execution while remaining under human strategic oversight.

This layer ensures that AI adoption is seen not as a set of isolated projects, but as part of a deliberate, phased organisational change. It helps leadership to get a clearer view of the whole journey and to set more realistic expectations around maturity levels and investment required.

Layer 2 – Process Readiness Layer

The middle layer is about the readiness of the whole process. It is mainly based on the process-oriented framework of Schmidt et al. [18], which considers five perspectives of process readiness:

- Activities - The nature and complexity of tasks performed.
- Decisions - Where authority lies and how decisions are made.
- Data Operations - Quality, availability, and flow of data.
- Control Flow - Sequencing, rules, and exception handling
- Resource Management - Allocation and coordination of people, systems, and tools.

This layer is based on the assessment of Process Debt, the mismatch that have developed between formally documented processes and the reality of daily practice. By comparing theoretical potential with real-world constraints, the layer produces a Practical Readiness score. This helps highlight hidden complexities and effort requirements early in the evaluation, directly supporting the broader goals of this thesis.

Layer 3 – Task Suitability Layer (Core Decision Tool)

Figure 5 illustrates how the bottom layer provides the most granular analysis. It employs Li and Ye's T-IPO model for task decomposition and the LARA matrix for readiness scoring [17]. The individual tasks are broken down into input, process, output, and constraints. The LARA matrix assesses them in several dimensions, with the focus on compliance sensitivity. This produces clear readiness levels (L1 to L4), which translate into practical deployment recommendations, from full agentic autonomy to human-led execution with AI support.

This layer is especially important for determining whether a specific task truly requires the reasoning and adaptability of agentic AI or can be adequately handled by simpler automation.

Integration Logic and Flow

The real strength of V-HAARF lies in the way its three layers work together. I designed the framework to support both top-down and bottom-up analysis.

In practice, the evaluation usually starts at the strategic level. This sets the wider organisational context and maturity. It then moves to the process level, where I assess structural readiness and process debt. Finally, it goes down to the task level for more specific recommendations. Insights can also go upwards. For example, recurring problems identified at the task level may be a signal of the need for strategic realignment or process redesign.

This interconnected structure leads to more holistic decisions. The framework not only generates a go/no-go recommendation, but also prioritizes actions, realistic effort estimates, and very clear guidance as to the domain knowledge that needs to be developed.

By combining strategic vision, process understanding, and task-level precision, V-HAARF offers Volvo a balanced and actionable tool. It supports consistent and well-informed AI adoption decisions across a wide range of business processes.

4.4 AI Environments

The Volvo Hybrid AI Adoption and Readiness Framework (V-HAARF) is designed as a general-purpose tool. It can be applied across many different business processes at Volvo Cars.

I deliberately avoided limiting it to one specific domain. Instead, the framework offers a flexible structure that works in any area where the company is exploring the use of agentic AI or related automation technologies [16], [18].

Applicability Across Volvo Processes

V-HAARF can support evaluation in diverse operational and engineering environments [17], including:

- Product development and verification processes
- Diagnostics and fault analysis systems
- Virtual simulation and testing environments
- Procurement and supply chain operations
- Compliance and regulatory reporting workflows
- Manufacturing support and quality control processes
- After-sales service and predictive maintenance
- Data management and reporting activities

Environments Where the Framework is Particularly Powerful

The framework delivers the greatest value in situations where decision-making is complex, and the choice between agentic AI and traditional automation is not obvious. It performs especially well in the following types of environments:

- **Complex and Data-Intensive Processes**: Where large volumes of data must be analysed and interpreted under varying conditions [18].
- **Regulated and Compliance-Heavy Domains**: Where decisions carry significant safety, legal, or quality risks, requiring strong governance and auditability [17].
- **Processes with Mixed Task Types**: Containing both deterministic (rule-based) tasks suitable for automation and interpretive or reasoning-heavy tasks that may benefit from agentic AI [3].
- **Legacy Systems and High Process Debt Environments**: Where existing processes have accumulated inconsistencies, undocumented variations, or fragmented data flows [18].
- **Knowledge-Intensive Workflows**: Where significant domain expertise is required, and the effort to develop this knowledge for AI systems needs careful estimation [16].

By supporting evaluation across these varied environments, V-HAARF helps Volvo make more strategic and risk-aware decisions about AI adoption. It encourages a balanced view, recognising that not every process requires advanced agentic capabilities, while identifying those where such systems can deliver substantial business value.

4.5 Types of Agents and Agent Design Patterns

A central consideration in the Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF) is determining the most suitable type of intelligent system for each process or task. The framework recognises that agentic AI exists on a spectrum of autonomy, and selecting the right level is crucial for achieving good business value while managing risk and implementation effort [3], [8].

Types of Agents

The framework defines four main types of agents based on their degree of autonomy and human involvement:

- **Assistive Agents**: Such systems attempt to assist users by drafting, suggesting, or automating simple repetitive tasks. Human judgment remains central to the outcome. They are suitable for processes where full autonomy is not yet justified or required.
- **Co-pilot Agents**: These agents work in real time with humans, analyzing information, making recommendations, and taking care of routine elements, while the human remains in charge and accountable. This type is particularly valuable in complex or knowledge-intensive domains [17].
- **Orchestrating Agents**: They can autonomously perform and coordinate several steps or sub-tasks to achieve stated goals. The human role is reduced to defining objectives and handling exceptions. They are appropriate for more mature processes with clear boundaries and governance structures.
- **Integrated Human-AI Systems**: In the long term, this is the ideal state for many processes with humans and AI agents working as a seamless team where AI handles execution and optimisation, and humans handle strategic direction, creativity, and ethical oversight [3], [16].

Agent Design Patterns

To ensure responsible and effective implementation, the framework recommends several key design patterns:

- **Human-in-the-Loop Governance**: Important decisions, especially those involving safety, compliance, or high uncertainty, should always include human review and approval.
- **Goal-Oriented Design**: We should not give agents a fixed set of instructions on how to do things but give them well-defined goals and constraints within which they can adapt.
- **Continuous Feedback Loops**: Mechanisms should be established so that agents can learn from human corrections and improve performance over time.

- **Explainability and Transparency**: All significant agent actions should be logged and explainable to support trust, auditability, and regulatory compliance.
- **Escalation Protocols**: Agents must automatically escalate issues to humans when predefined risk or uncertainty thresholds are exceeded.

These agent types and design patterns help Volvo implement AI solutions in a controlled, scalable, and responsible manner. They support the scope by ensuring that decisions about AI adoption are matched to the specific characteristics and requirements of each process.

5 Implementation Method

This chapter explains how teams at Volvo Cars can systematically apply the Volvo Hybrid AI Adoption and Readiness Framework (V-HAARF) to evaluate business processes throughout the company. I designed the approach to be repeatable, scalable, and flexible enough to work across different departments. It supports the company's broader goal of making consistent and well-informed decisions about where and how to adopt agentic AI.

5.1 Method of Applying the Framework

The V-HAARF framework was developed in an iterative, mixed-methods approach based on Design Science Research (DSR) [19], which offers sufficient structure for consistency, but sufficient flexibility to adapt to different contexts.

It ensures that both strategic considerations and real operational realities are carefully considered when determining whether a given process should adopt agentic AI, rely on traditional automation, or remain primarily under human control.

The implementation follows five main iterative steps:

1. Process Mapping and Task Decomposition
First, the selected process is described in detail. With the help of the T-IPO model, complex activities are decomposed into individual tasks with clearly defined inputs, processes, outputs, and constraints. This step establishes a solid understanding of the process structure and helps identify areas requiring interpretation versus rule-based execution [17].

2. Multi-Stakeholder Workshop for Scoring
Cross-functional workshops are conducted involving process owners, domain experts, IT specialists, and compliance representatives. Participants evaluate the process using Schmidt et al.'s five perspectives (Activities, Decisions, Data Operations, Control Flow, and Resource Management) and assess process debt. Initial readiness scoring is performed collaboratively to incorporate diverse viewpoints [18].

3. Quantitative Scoring and Qualitative Validation
The LARA matrix is intended to produce numerical readiness scores at the task level, with a focus on compliance sensitivity. These scores are combined with qualitative insights from workshops to validate findings. This step produces a balanced view of both potential and practical readiness, including estimates of required time, resources, and domain knowledge development.

4. Pilot Deployment on High-Readiness Tasks
Controlled pilot implementations choose high-readiness-score tasks or subprocesses. This step tests the performance of selected agent types (assistive, co-pilot, or orchestrating) in the real world and helps refine the framework based on real-world outcomes and user feedback.

5. Continuous Recalibration and Improvement
Pilot results, evolving business needs, and advances in AI result in periodic updates to the framework. Readiness scores, process debt levels, and decision criteria are recalibrated to maintain relevance over time.

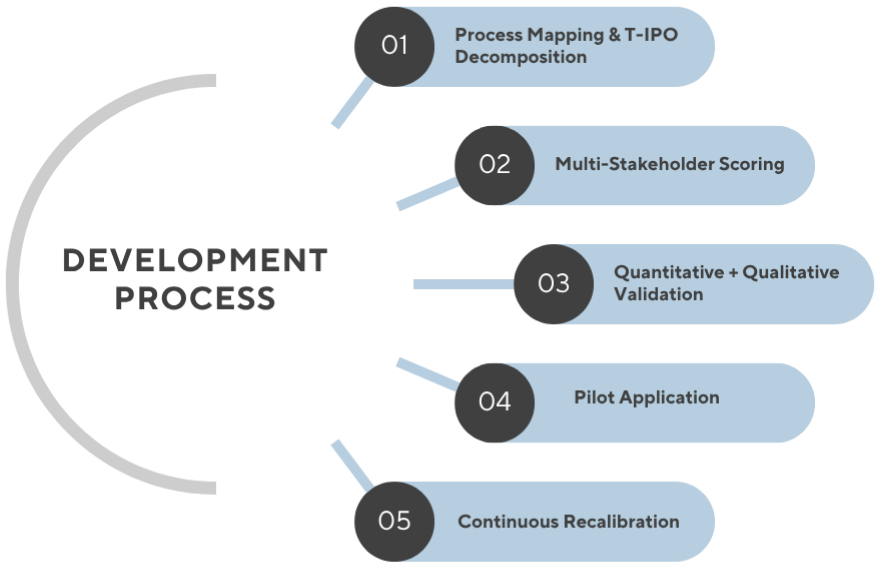

Figure 6 Iterative Development Process of the V-HAARF Framework

Figure 6 outlines the practical five-step process for applying V-HAARF to any business process at Volvo Cars. This five-step approach provides a comprehensive, evidence-based, and flexible evaluation. It directly addresses the increased scope by providing a simple way to evaluate feasibility, business value, implementation difficulty, and best practices for any process at Volvo Cars. By following this structured approach, teams can move from initial process selection to actionable recommendations with transparency and confidence.

5.2 Best Practices for Implementation

Successful application of the Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF) depends not only on following the method but also on observing proven best practices. These practices have been derived from the synthesis of recent literature and are tailored to support Volvo's expanded need for systematic, responsible, and value-driven AI adoption decisions across diverse business processes.

The following best practices are recommended when using the framework:

- **Begin with Strategic Alignment**: Always start the evaluation at Layer 1 (Strategic Evolution) to ensure leadership commitment and organisational readiness before investing resources in detailed assessments. Premature focus on individual tasks often leads to misaligned efforts [16].
- **Conduct Thorough Process Debt Assessment**: Before considering agentic AI, rigorously identify and quantify process debt at Layer 2. Processes with high debt

should first be stabilised and documented, as introducing AI into chaotic environments significantly increases risk and effort [18].

- **Insist on Task-Level Granularity**: Never base decisions solely on high-level activities. Use Layer 3 (T-IPO decomposition and LARA matrix) to analyse individual tasks. This prevents over-estimation of AI suitability and supports accurate differentiation between rule-based automation and agentic AI needs [17].
- **Apply a Non-Compensable Compliance Floor**: In regulated or safety-critical processes, compliance sensitivity must be treated as a strict requirement. Tasks failing minimum compliance thresholds should not proceed to agentic AI deployment, regardless of other favourable scores [17].
- **Realistically Estimate Effort and Domain Knowledge Needs**: Explicitly calculate the time, resources, and knowledge development required at each layer. This includes both technical integration effort and the human expertise needed to train, supervise, and maintain the systems [3].
- **Use Pilot Deployments with Strong Governance**: Run small-scale pilots of high-readiness tasks (L1 or L2). Use tight governance, monitoring, and human oversight during pilots to gather real performance data before scaling up [8], [18].
- **Enable Continuous Recalibration**: Treat the framework as a living tool. Periodically update readiness scores, LARA weights, and decision criteria as AI technologies evolve and organisational maturity increases [17].
- **Promote Cross-Functional Collaboration and Knowledge Transfer**: Involve stakeholders from business, IT, compliance, and operations throughout the process. Document lessons learned and store them systematically to support knowledge transfer between projects [16].
- **Maintain Transparency and Auditability**: Thoroughly document all evaluations, scoring decisions, and recommendations. This supports internal governance and builds trust in the decision-making process.

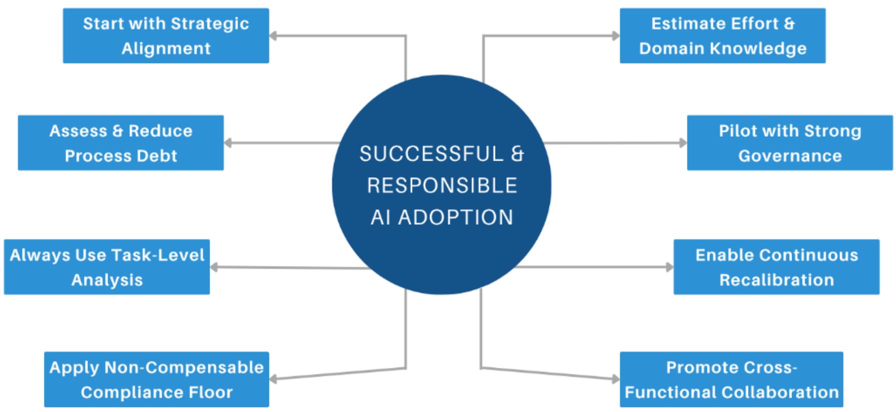

Figure 7 Key best practices for implementing the Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF)

Figure 7 highlights the most important best practices identified during the development and application of V-HAARF. By following these best practices, highlighted in Figure 7, Volvo can mitigate risks, optimise resource allocation, and improve the likelihood of successful AI initiatives. They transform V-HAARF from a theoretical model into a reliable operational tool for responsible AI adoption.

5.3 Adaptation Instructions for Different Volvo Domains

The Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF) is intentionally designed to be adaptable across the wide range of business processes within Volvo Cars. Adaptation starts by analysing the specific characteristics of the target domain, including its risk level, regulatory requirements, data complexity, and strategic importance [16], [18].

The importance of each layer should be scaled according to the nature and risk profile of the domain in the application of the Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF). The role of processes in the company such as test planning and allocation in the support of vehicle safety verification and regulatory certification is critical. Although these processes are planning and coordination activities, they come with significant compliance and safety-related risks. As a result, more weight should be given to Layer 1 (Strategic Evolution) and the governance aspects of Layer 2 (Process Readiness) [17]. This ensures strong alignment with organisational policies, risk tolerance and compliance requirements 18].

The relative weighting of the three layers remains adjustable. If the overall structure stays intact, this flexibility keeps V-HAARF relevant and effective across different contexts, from software-intensive engineering processes to data-heavy operational workflows.

5.4 How the V-HAARF Framework Helps Volvo Cars

The primary goal of the development of the Volvo Hybrid AI Adoption and Readiness Framework (V-HAARF) was to provide Volvo Cars with a practical and strategic tool for responsible AI adoption. The framework offers a systematic approach to measuring processes and, in doing so, introduces several tangible benefits that directly contribute to the company's long-term aspirations in digital transformation.

First, V-HAARF offers a standardised and repeatable evaluation process. It can be used in a consistent manner across any department or business area by teams. This reduces inconsistency in decision-making and ensures that every AI initiative is assessed against the same clear criteria, no matter the domain [16], [18].

Second, the framework enables more informed and data-driven decisions to be made when selecting between agentic AI, traditional automation, and human-driven operations. Its multi-level analysis, strategic, process, and task, carefully evaluates key factors such as data flow characteristics, the balance between fixed rules and tasks requiring interpretation, compliance demands, and the level of process debt. This assists teams in avoiding deploying advanced AI where it is unlikely to succeed and identifies processes where there is a high probability of a meaningful improvement [3], [17].

Third, the framework helps reduce the risk of wasted investment. Surfacing high process debt or low readiness early in the evaluation, it prevents teams from launching overly ambitious AI projects that are likely to underperform or demand far more resources than anticipated [18].

Fourth, the framework permits more realistic planning. It explicitly accounts for the time, resources, and domain-specific knowledge necessary to successful implementation, directly addressing one of the most common pitfalls in AI adoption: underestimating the real effort involved. This allows the teams to formulate more robust business cases and project plans [8], [16].

Fifth, the framework supports knowledge transfer and organisational learning. Its structured outputs, scoring mechanisms, and supporting tools like evaluation templates and checklists embed reusable assets. These can be shared across projects and departments, helping future AI initiatives get off to a stronger start and move forward more quickly [17].

Finally, V-HAARF upgrades the profile as a more sophisticated and strategic user of AI. Rather than jumping on the bandwagon of technology hype, the company can show a purposeful, evidence-led attitude to intelligent automation. This strengthens governance, builds greater stakeholder confidence, and supports a more sustainable long-term digital transformation.

Overall, the framework shifts AI adoption from an often activity into a well-managed, company-wide capability when necessary. It offers Volvo both practical decision support and a solid foundation for continuous improvement in how intelligent technologies are evaluated and implemented across the organisation.

6 A Post-Implementation Evaluation Framework for Agentic AI Systems

While the Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF) focuses on the strategic decision of whether agentic AI should be adopted for a given process, organisations also need robust tools to evaluate AI systems after they have been implemented or piloted. This chapter introduces a complementary focused evaluation framework designed specifically for post-implementation assessment. It serves as a supporting instrument within the broader V-HAARF approach.

6.1 Rationale and Motivation

Assessment needs to continue even after a decision has been made to deploy agentic AI. Barriers to many AI initiatives include real-world performance, integration issues, governance gaps, user uptake, and long-term sustainability. Without a structured post-implementation evaluation method, it is difficult to measure real business value, identify improvement areas, justify continued investment, or decide whether to scale the solution.

This focused framework was developed to address these needs. It provides Volvo with a practical tool to systematically assess the effectiveness, maturity, and organisational impact of deployed agentic AI systems, thereby supporting responsible and sustainable AI adoption across the company.

6.2 Theoretical Foundations

This post-implementation evaluation framework draws on three complementary perspectives from recent literature [6], [7], [8]:

- One perspective emphasises lifecycle awareness, governance, auditability, knowledge transfer, and long-term sustainability.
- Another provides a developer-centric evaluation across five key dimensions, including learning cost, functional abstraction, development efficiency, performance optimisation, and maintainability.
- The third focuses on multi-faceted real-world deployment aspects such as accuracy, robustness, cost-effectiveness, user experience, and human-AI collaboration quality.

These perspectives were selected because they collectively cover both governance-related and practical performance-related aspects of deployed AI systems, creating a balanced evaluation approach.

6.3 Overall Architecture of the Post-Implementation Evaluation Framework

This post-implementation evaluation framework in Figure 8 complements the main V-HAARF framework. While V-HAARF supports the initial decision of whether to adopt agentic AI, this framework helps monitor, validate, and continuously improve AI systems once they are in active use.

- **Layer 1 – Lifecycle and Governance Foundation**

This layer evaluates the system's alignment with organisational governance standards, auditability, compliance requirements, knowledge transfer mechanisms, and long-term sustainability [7].

- **Layer 2 – Developer and Maintainability Perspective**

This layer assesses the internal quality and long-term viability of the AI system from a development and technical perspective. It examines factors such as learning cost, functional abstraction, development efficiency, performance optimisation, and maintainability [6].

- **Layer 3 – Real-World Performance and Impact**

This layer focuses on the practical outcomes and business value delivered by the AI system under real operating conditions. It measures accuracy, robustness, user experience, cost-effectiveness, and the quality of human-AI collaboration [8].

Figure 8 Post-Implementation Evaluation Framework for Agentic AI Systems

Figure 8 presents the overall architecture of the post-implementation evaluation framework applied during this thesis. The framework integrates three complementary layers: Lifecycle & Governance, Developer Experience & Maintainability, and Real-World Performance & Impact.

6.4 Scoring Concept

The framework uses a transparent weighted scoring system to combine the outputs of the three layers into one overall AI System Effectiveness Score (ranging from 0 to 100). The formula is defined as:

AI System Effectiveness Score = (0.30 × Layer 1) + (0.40 × Layer 2) + (0.30 × Layer 3)

Each layer is scored on a scale from 1 to 5, where:

- 1 = Very low / Poor performance (significant deficiencies, high risk, or major gaps)
- 2 = Low (noticeable weaknesses that require substantial improvement)
- 3 = Moderate / Acceptable (basic functionality is present, but clear room for improvement)
- 4 = Good / Strong (solid performance with only minor limitations)
- 5 = Excellent / Very strong (high maturity, minimal issues, and clear added value)

These criteria provide clear guidance so that scoring remains evidence-based rather than subjective.

For illustrative purposes, the framework was conceptually applied to an ongoing AI initiative within Volvo Cars. According to this application, the evaluated system received an overall AI System Effectiveness Score of 73 out of 100. This score indicates that while the system demonstrates promising capabilities, there are identifiable areas for improvement, particularly in response latency, edge-case handling, and certain governance aspects. This example demonstrates how the framework can produce concrete and interpretable results to support decision-making [3].

6.5 Practical Application Guidelines

The three-layer post-implementation evaluation framework can be used at natural stages of the AI system lifecycle, e.g. post-pilot stages, post-major upgrades or at regular evaluation points of live AI projects. The guidance emphasizes the importance of cross-functional collaboration, clear documentation and continuous improvement and offers practical tips on how to adapt the weights and metrics to the specific risk level and strategic importance of each process. This allows for flexibility, while maintaining consistency between evaluations.

Overview of Identified Issues (Categorisation and Frequency)
When applying the framework, I first identify and categorise the issues that arise during the operation of the AI system in a systematic manner. I categorise the issues according to the three layers: governance and lifecycle issues, developer and maintainability issues, and performance issues in the real-world. Particularly useful was the recording of the frequency of each of the issues. This helped reveal those problems occurring repeatedly and enabled teams to prioritise the areas needing most attention [6].

Template Analysis (First-Order Codes, Second-Order Themes, Aggregate Dimensions/CSFs)
I employed Template Analysis, as shown in Table 1, to identify significant patterns in the data I collected. The process started with first-order codes that captured specific

observations from the evaluation. These were then grouped into broader second-order themes and finally organised into aggregate dimensions that align with the three layers of the framework. This structured approach brought important insights. It highlighted critical success factors and made both the systemic strengths and weaknesses of the AI system's deployment much clearer [8].

Table 1. 2nd-order Themes and Aggregate Dimensions from Template Analysis

2nd-order Theme	Aggregate Dimension	Description
Consistency & Reliability	Technical Performance & Reliability	AI produces correct, repeatable results across operations (split/merge, move, scale, replot, sync)
Data Integrity & Reliability	Technical Performance & Reliability	Prevention of data loss, duplication, or corruption during modifications
System Stability & Performance	Technical Performance & Reliability	Server uptime, response time, and overall robustness under load
Usability & User Experience	User-Centric Design & Interaction	Interface clarity, visual layout, ease of interaction, and intuitive feedback
Explainability & Feedback Issues	User-Centric Design & Interaction	Quality of AI explanations, feedback options, and learning from user input
AI Limitations & Hallucinations	AI Maturity & Continuous Learning	System errors, server issues, unexpected behaviour, and capability gaps

Effort, Time, and Complexity Analysis
A detailed effort and complexity analysis determines the resources necessary to cover the identified flaws, including the time and people needed for governance improvements, technical refactoring, testing and knowledge development. The analysis offers realistic projections that can serve for planning and resource allocation [6].

Performance and Validation Results
The last step is to measure the performance of the system against the defined criteria on each layer. Layer 1 is governance and compliance, layer 2 is maintainability and quality of development and layer 3 is real-world impact and user experience. These results are combined using the weighted scoring formula to produce an overall AI System Effectiveness Score, offering clear evidence of current strengths and limitations to guide future decisions [8].

6.6 Step-by-Step Evaluation Process

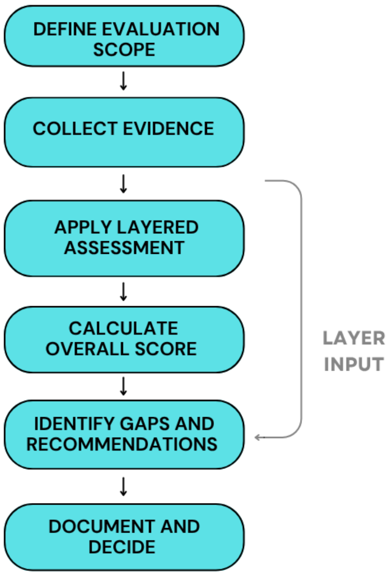

Figure 9 Step-by-step evaluation process of Post-Implementation Evaluation Framework for Agentic AI Systems

Figure 9 outlines the practical step-by-step process for applying the post-implementation evaluation framework. The process started with a clear definition of the scope of the evaluation and proceeded with the collection of evidence, layered assessment, calculation of scores, identification of limitations, and documentation of the results. This systematic approach resulted in a systematic, transparent, and repeatable evaluation.

The practical evaluation follows a clear six-step process:

- Define Evaluation Scope - Clarify the objectives, boundaries, and success criteria for the assessment.
- Collect Evidence - Gather performance data, user feedback, development logs, and operational metrics.
- Apply Layered Assessment - Systematically evaluate the solution across the three layers (Governance & Lifecycle, Developer & Maintainability, and Real-World Performance).
- Calculate Overall Score - Combine the layer scores using the predefined weighting to generate the AI System Effectiveness Score.
- Identify Gaps and Recommendations - Highlight improvement areas and propose specific, prioritised actions with estimated effort.
- Document and Decide - Produce a concise report with a clear recommendation (continue, enhance, pause, or decommission) and archive insights for organisational learning.

This structured process, presented in Figure 9, highly supports that evaluations are thorough, reproducible, and directly connected to business value and risk management [8],

[18]. By offering both a scoring mechanism and a practical protocol, this post-implementation framework helps Volvo maintain visibility, control, and continuous improvement over its agentic AI initiatives.

6.7 Adaptation Guidelines for Different Volvo Domains

The framework is intentionally domain-agnostic, allowing it to be used in a wide range of Volvo processes such as product development, virtual simulation, diagnostics, manufacturing support, procurement, compliance reporting, and after-sales service. Adaptation begins by identifying the specific characteristics of the target domain, including its risk level, regulatory requirements, data complexity, and strategic importance [6], [8].

For domains with high regulation, e.g., verification or compliance procedures, I propose to put more focus on Layer 1 (Lifecycle and Governance). This means higher compliance thresholds and more rigorous audit requirements [7]. On the other hand, operational efficiency domains (e.g., manufacturing support, procurement) are more suited to put more weight on Layer 3 (Real-World Performance), where the assessment can be based on quantifiable results such as cost reduction and process speed [8].

The relative weights of the three layers can be adjusted as needed, if the overall structure is preserved. This flexibility enables the framework to be relevant and efficient in the case of software-heavy engineering processes or data-heavy operational workflows [6].

6.8 Reusable Tools, Templates, and Supporting Materials

This set of practical, lightweight tools is designed to enable people to use the framework in a consistent and effective manner. These tools are designed to reduce manual effort, improve repeatability, and support knowledge transfer between projects. They are intentionally built in commonly available formats (primarily Excel) to ensure easy adoption across different Volvo teams [1].

As practical outcomes of the framework development, the following supporting tools were created:

- Evaluation Scoring Template
- Issue Taxonomy and Checklist
- Effort and Complexity Estimation Model
- Governance and Risk SLO Checklist

6.9 Evaluation Scoring Template

The Evaluation Scoring Template is a Microsoft Excel based tool which automatically calculates the AI System Effectiveness Score. It includes individual sheets for all three layers of the score, with fields to enter scores and supporting evidence for governance, developer-centric dimensions, and real-world metrics. The template automatically applies the predefined weights and produces an overall score, along with visual indicators (radar charts and summary dashboards) to highlight strengths and improvement areas [1], [2].

Issue Taxonomy and Checklist

The Issue Taxonomy and Checklist classify the frequent issues arising in AI system evaluations into structured categories, assisting teams in systematically capturing issues related to performance, maintainability, governance and usability. The checklist consists of pre-defined questions mapped to the three layers, ensuring that no critical aspect is missed during evaluations [2].

Effort and Complexity Estimation Model
The model offers a systematic approach to estimating time, resources and effort required to close the identified gaps It breaks down effort by phases (e.g. governance improvements, technical refactoring, testing, knowledge development) and applies scaling factors based on process complexity and domain risk level. The model supports realistic planning and helps justify resource allocation for AI system enhancements [2].

Governance and Risk SLO Checklist
The Governance and Risk SLO (Service Level Objective) checklist makes sure that the assessed AI systems meet Volvo's internal requirements on compliance, auditability, transparency and risk management. It contains clear acceptance criteria and recommended mitigation actions for common governance risks, making it particularly useful for regulated domains [3].

These tools transform the theoretical framework into a ready-to-use operational kit. By providing standardised templates and checklists, they promote consistency, reduce duplication of effort, and support knowledge sharing across different Volvo initiatives [1].

7 Results

The V-HAARF framework presented in Chapter 4 remains a conceptual design. In contrast, the post-implementation evaluation framework was empirically applied to the CTPBot project, as described in this chapter. This real-world application allowed validation of the scoring method and visualisation approach using actual project data, including logged issues, effort records, and validation reports.

This chapter presents the application of the three-layer post-implementation evaluation framework to the ongoing CTPBot project in the company. The purpose is to demonstrate how the framework works in practice when evaluating an already implemented agentic AI system.

Overview of the CTP Process and CTPBot Solution
The CTP process is an important planning tool at Volvo Cars for distributing the test activities to the available test vehicles. The CTPBot was introduced as an agentic AI solution to support and automate parts of this planning workflow. The system assists with tasks such as plan generation, test need modifications (move, split, merge), car removal, and alternative proposals. As the bot was under active development and pilot usage during the thesis period, it provided an ideal real-world case for applying the post-implementation evaluation framework.

7.1 Empirical Data Collection

The evaluation was based on comprehensive empirical data gathered over the thesis period. This included 42 detailed issue logs captured during scenario-based testing, weekly effort logs recording time spent on evaluation activities, and supplier-provided validation reports containing 149 test cases. These data sources offered concrete evidence of system behaviour, user interactions, and performance needs [6], [8].

7.2 Mapping Between Layers in The CTP Workflow

The three-layer framework was mapped directly to the CTPBot system. Layer 1 (Lifecycle and Governance) considered auditability, knowledge transfer mechanisms, and compliance aspects [7]. Layer 2 (Developer and Maintainability) evaluated technical quality, maintainability, and development efficiency. Layer 3 (Real-World Performance and Impact) evaluated accuracy, response latency, usability, and overall value delivered to planners. This mapping allowed a structured analysis tailored to the actual CTP workflow.

7.3 Scoring and Visualisation Approach

Each layer was scored from 1 to 5 based on defined criteria, and the scores were combined using the weighted formula to calculate the overall AI System Effectiveness Score. Figure 10 illustrates a dashboard concept that was used to visualise the results, including radar charts for Layer 2 dimensions and summary metrics from all three layers. This provided a clear, at-a-glance view of the system's current strengths and weaknesses.

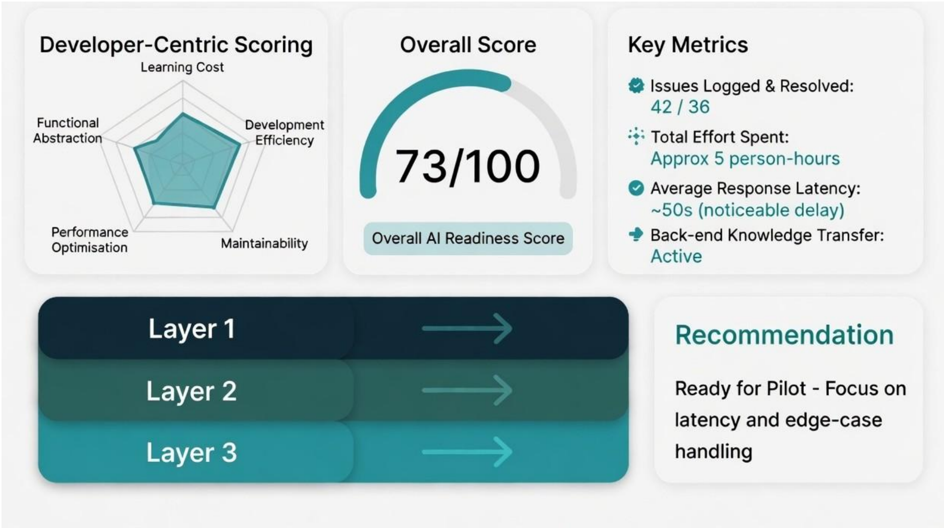

Figure 10 A Dashboard Concept of AI System Effectiveness Score

Figure 10 shows the dashboard-style visualisation developed to present the results of the post-implementation evaluation. It combines the overall AI System Effectiveness Score, a radar chart showing performance across the five developer-centric dimensions, important metrics such as logged issues, effort spent, and response latency, and a clear recommendation for next steps. This type of visualisation makes the evaluation results easy to understand for both technical teams and management.

It is scored 73/100 on the AI System Effectiveness Score. This shows it is promising and partially ready for wider adoption, and needs further improvements, especially for response latency and edge-case management.

7.4 Practical Application Guidelines and Step-by-Step Usage Protocol

The framework was applied following a six-step protocol: defining the evaluation scope, collecting evidence, performing a layered assessment, calculating the overall score, identifying gaps with recommendations, and documenting findings for organisational learning. This structured process ensured the evaluation was systematic and repeatable.

Evaluation Scoring Template

To assist the evaluation, a scoring template was created in Excel to record the scores as well as the evidence for each layer. The weighted overall score was then automatically calculated and visual summaries were produced, thus enabling an efficient and consistent evaluation.

Issue Taxonomy and Checklist

Table 2 presents that all 42 logged issues were categorised using the Issue Taxonomy and Checklist. Issues were grouped under the three layers, allowing clear identification of recurring themes related to governance, maintainability, and real-world performance.

Table 2. Issue Taxonomy Derived from the CTPBot Evaluation

Phase	Average Hours	Percentage of Total Effort	Notes / Scaling Guidance	Volvo Recommendation
Consistency & Reliability	13	31 %	Technical Performance & Reliability	Implement automated post-operation consistency checks
Usability & User Experience	10	24 %	User-Centric Design & Interaction	Standardise UI feedback patterns and error recovery
AI Limitations & Hallucinations	9	22 %	AI Maturity & Continuous Learning	Strengthen prompt engineering and feedback loops
Explainability & Feedback Issues	4	10 %	User-Centric Design & Interaction	Improve free-text handling and feedback options

Effort and Complexity Estimation Model
The Effort and Complexity Estimation Model shown in Table 3, was applied to quantify the resources needed to address the identified issues. It estimated effort across phases such as governance improvements, technical fixes, testing, and knowledge development, providing realistic projections for future refinement of the system.

Table 3. Effort and Complexity Estimation Model (based on CTPBot evaluation)

Phase	Average Hours	Percentage of Total Effort	Notes / Scaling Guidance
Onboarding & Document Review	30	8 %	One-time setup per project
Scenario-based Testing	140	37 %	Scales with number of use cases
Issue Logging & Feedback	165	44 %	Highest effort phase
Validation & Refinement	30	8 %	Can be reduced with better automation
Template Analysis & Framework Application	30	8 %	One-time per major evaluation
Total	~395	100 %	Adjust using scaling factor for project size

7.5 Governance and Risk SLO Checklist

The Governance and Risk SLO Checklist was used as shown in Table 4 to verify compliance, auditability, and risk management aspects. It highlighted areas where stronger controls were needed to meet Volvo's internal standards.

Table 4. Governance and Risk SLO Checklist (Layer 1)

Item	Description / Requirement	Recommended Acceptance Criteria	Suggested Mitigation Action
Auditability of Decisions	All AI actions must be logged and traceable	Full log of every operation	Implement back-end logging
Knowledge Transfer & Reuse	Corrections and optimisations stored for reuse	>80% of issues stored and reused	Enable the back-end storage mechanism
Data Privacy & Compliance	No sensitive data exposed	Full compliance with Volvo policy	Apply data masking and access control
Response Latency	Acceptable delay for interactive use	< 3 seconds average	Optimise prompt engineering and model selection
Error Rate & Robustness	Acceptable failure rate	< 5% critical errors	Add fallback mechanisms and user guidance
Overall Risk Level	Combined risk assessment	Low / Medium	Conduct a formal risk review before the pilot

The three-layer post-implementation evaluation framework used for the CTPBot project provided an overall AI System Effectiveness Score of 73 out of 100, which is a promising level of performance, and indicates that the system already delivers useful functionality in support of the CTP workflow. The structured evaluation also revealed specific opportunities for further refinement, regarding response latency, edge-case handling, and governance aspects.

Importantly, the framework provided clear, evidence-based insights that complement the development work already carried out by the supplier team. These insights offer constructive guidance for prioritising future improvements and strengthening the overall solution.

8 Analysis and Discussion

This chapter analyses the two frameworks developed in this thesis. The main Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF) and the complementary three-layer post-implementation evaluation framework. It examines their feasibility and business value, the effort and complexity involved in their application, the methods and best practices that emerged, and how they align with current literature. The discussion focuses on their potential contribution to Volvo Cars' strategic goal of making informed and consistent decisions about agentic AI adoption across different business processes.

8.1 Feasibility and Business Value of the Frameworks

Both frameworks show a good viability for industrial use. V-HAARF provides a structured, multi-level approach that enables Volvo to evaluate any existing process and decide whether agentic AI, traditional automation, or human-led operations would be most appropriate. Its three-layer design (strategic, process, and task) makes it practical and scalable across departments. The framework's business value lies in its ability to reduce wasted investment by identifying unsuitable processes early and supporting realistic go/no-go decisions.

The complementary three-layer post-implementation evaluation framework adds further value by allowing Volvo to assess AI systems that are already in operation or pilot stage. It delivers measurable insights into governance, maintainability, and real-world performance, helping the company continuously improve deployed solutions. Together, the two frameworks create a complete pathway, from initial process evaluation to ongoing system assessment, which significantly strengthens Volvo's AI adoption capability [16], [18].

8.2 Time, Resources, and Effort Required for Domain-Specific AI Knowledge

One of the key findings is that building enough domain specific knowledge is still one of the most time consuming and resource intensive parts of AI adoption. V-HAARF includes effort estimation at each layer enabling teams to plan for the real costs of alignment, process debt reduction and task level improvements. The post-implementation framework also emphasizes that even after deployment there is an ongoing effort needed for governance upkeep, model recalibration, and knowledge transfer. These insights confirm that realistic effort planning is essential for successful AI initiatives and should be a core part of any adoption decision [7], [17].

8.3 Methods, Best Practices, and Lessons Learned

The development process revealed several important methods and best practices. The iterative Design Science Research (DSR) approach proved effective for creating both frameworks. Key lessons include the importance of combining strategic, process, and task-level analysis rather than relying on any single perspective.

Best practices that emerged are:

- Always begin with strategic alignment before detailed evaluation.
- Systematically identify and address process debt before considering agentic AI.
- Use task-level granularity (T-IPO + LARA) for accurate AI vs automation decisions.
- Apply weighted multi-layer scoring to balance governance, maintainability, and real-world impact.
- Treat evaluation as an ongoing process rather than a one-time activity.

These practices provide Volvo with actionable guidance for future AI projects [17], [18].

8.4 Alignment with Existing Literature and Theoretical Frameworks

The two frameworks are in line with recent studies, while extending the studies to the industrial context of Volvo. V-HAARF combines the strategic evolution model by Ragsdale, the process-oriented readiness assessment by Schmidt et al., and the task-level analysis by Li & Ye, into a single and coherent tool. The post-implementation framework draws on Farooq et al.'s governance focus, Wang et al.'s developer-centric evaluation, and Myakala's multi-faceted real-world metrics. This synthesis addresses a limitation in the literature, where few studies combine strategic, process, and task-level perspectives into practical, company-wide frameworks [6], [7], [18].

8.5 Contribution and Implications for Volvo Cars

The main contribution of this thesis is two complementary frameworks that together supported the entire AI adoption lifecycle at Volvo Cars. V-HAARF supports the company in making better initial decisions while the post-implementation framework supports continuous evaluation and improvement of deployed systems. These tools can help Volvo adopt AI more strategically, reduce unnecessary risk and cost, and build a mature, systematic approach to intelligent automation.

9 Conclusion

This thesis set out to address thesis scope for structured support in evaluating where and how agentic AI should be introduced across its business processes. In response to the thesis scope, two complementary frameworks were developed: the Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF) as the primary decision-making tool, and a focused three-layer post-implementation evaluation framework to assess AI systems after deployment. Together, these frameworks provide Volvo with guidance across the full lifecycle of AI adoption from initial suitability assessment to ongoing performance evaluation.

9.1 Summary of Key Findings

The main contribution of this thesis is the design of V-HAARF, a multi-level framework to evaluate any business process to determine if it is suitable for agentic AI, traditional automation, or for continued human-led operations. The framework combines strategic, process, and task-level perspectives, with practical tools for effort estimation and decision-making.

Moreover, a complementary three-layer post-implementation evaluation framework was proposed to evaluate existing AI systems considering governance and lifecycle aspects, developer and maintainability dimensions, and real-world performance, providing a structured way to measure effectiveness after deployment. The study also produced a set of reusable tools and templates, along with clear methods and best practices for applying both frameworks in an industrial context.

9.2 Contributions to Volvo Cars and Academic Knowledge

This thesis contributes in two important ways. For Volvo Cars, it provides a practical, reusable approach to support strategic decision-making around the adoption of agentic AI. V-HAARF enables the company to assess feasibility, business value, and effort needed before undertaking AI projects, and the post-implementation framework supports continuous assessment and improvement of deployed solutions. These tools can support Volvo in making AI adoption more systematic and reducing the risk of less prepared implementations.

From an academic perspective, the thesis contributes by consolidating and extending existing models into two coherent, practice-oriented frameworks. V-HAARF combines strategic evolution perspectives with process readiness and task-level analysis, and the post-implementation framework synthesises governance, developer-centric and real-world evaluation dimensions. This dual-framework approach addresses a gap in current research, where few studies provide structured support across both the pre-adoption and post-implementation phases of agentic AI in industrial settings.

9.3 Limitations of the Study

There are few limitations of this study. Both frameworks were developed conceptually and have been validated primarily through literature synthesis and illustrative application rather than large-scale empirical testing across multiple processes. The post-implementation framework was applied to one ongoing project, which provided useful insights but limits the generalisability of the findings. Additionally, the short duration of the thesis constrained the depth of validation and the opportunity to test the frameworks in different organisational contexts.

10 Future Work

This thesis has developed two complementary frameworks to support Volvo Cars make more structured and informed decisions about the adoption and evaluation of agentic AI. While both frameworks provide a solid foundation, several promising directions remain for further development and validation. These future efforts can help strengthen the practical applicability and organisational impact of the work.

Empirical Validation of V-HAARF

One of the most important areas for future work is the empirical validation of the Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF). Although the framework was developed based on established literature and designed to be generally applicable, it has so far only been conceptually illustrated. Applying V-HAARF across multiple processes in different departments (such as manufacturing support, diagnostics, procurement, and compliance) would provide valuable insights into its usability, scoring reliability, and adaptability in real organisational settings.

Development of Digital Support Tools

Both frameworks rely on Excel-based templates, and the development of a more sophisticated digital tool, such as a web-based evaluation platform or integrated dashboard, would greatly improve usability, consistency, and scalability. Such a tool could automate scoring, produce visual reports, archive historical evaluations, and facilitate knowledge transfer across projects. This would make the frameworks more accessible for wider adoption across Volvo Cars.

Longitudinal Application of the Post-Implementation Framework

The three-layer post-implementation evaluation framework was applied to one ongoing AI initiative. Future research might include longitudinal studies where the same AI systems are subjected to the framework multiple times over time (e.g., every 6 or 12 months). This would enable researchers and practitioners to understand how the effectiveness of AI systems changes over time and how the framework's recommendations translate into real-world improvements.

Domain-Specific Customisation

I designed both frameworks to be domain-specific, but there is obvious room for future development of tailored versions or specific guidelines for different areas in Volvo Cars. For example, software development processes or supply chain processes could benefit from custom adaptations. Changing layer weightings or adding domain-specific criteria would probably improve the relevance and accuracy of evaluations in those contexts.

Integration with Existing Volvo Processes and Tools

Future work should also consider how these frameworks can be integrated more closely into existing Volvo project management, governance, and AI development processes. For example, understanding how V-HAARF can be effectively applied at the early stages of initiation of AI projects. Likewise, associating the post-implementation assessment framework with existing performance tracking and audit processes would facilitate easier integration with standard practices.

Extension to Emerging AI Technologies
With the fast pace of development of agentic AI and generative AI technologies, further research may extend the frameworks to cater for new challenges. This may include evaluation of multi-agent systems, evaluation of risks and opportunities of generative AI in engineering workflows as well as more explicit inclusion of ethical and sustainability dimensions into the evaluation criteria.

Organisational Adoption and Capability Building
Finally, future research could explore how Volvo Cars can develop internal capability to effectively use these frameworks. This could involve the development of training materials, running workshops and producing internal guidelines that would allow teams across the organisation to use the frameworks on their own. In addition, exploring the organisational adoption process itself would provide valuable insights into change management and knowledge transfer in relation to AI evaluation practices.

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A. Appendix

A.1 Supporting Materials for V-HAARF

Each layer in V-HAARF is evaluated using a 1 to 5 scale. The scoring criteria are defined as follows:

Score	Description	Interpretation
5	Excellent	Very high maturity, minimal gaps, strong strategic alignment
4	Good	Solid performance with only minor limitations
3	Moderate	Acceptable, but clear room for improvement
2	Low	Noticeable weaknesses that require significant improvement
1	Very Low	Major deficiencies, high risk, or fundamental gaps

Layer-wise Guidance:

- Layer 1 (Strategic Evolution): Evaluated based on leadership commitment, process visibility, data & policy alignment, and long-term vision.
- Layer 2 (Process Readiness): Assessed using the five perspectives (Activities, Decisions, Data Operations, Control Flow, Resource Management) and the level of process debt.
- Layer 3 (Task Suitability): Based on task decomposition using T-IPO and the suitability for agentic AI vs automation using LARA principles.

A.2 Effort and Complexity Estimation Model

This model provides indicative effort ranges (in person-days or person-weeks) for applying V-HAARF across different process sizes. It includes phases such as:

- Strategic Alignment
- Process Mapping & Debt Assessment
- Task-Level Analysis
- Pilot Planning & Recommendation Development

A.3 Adaptation Guidelines for Different Volvo Domains

V-HAARF, the Volvo Hybrid AI Adoption & Readiness Framework, is designed to be flexible and applicable for different business domains within Volvo Cars. However, different domains will have different risk, regulation, data complexity and strategic importance and therefore the framework should be adapted accordingly to ensure relevant and meaningful assessments.

This appendix provides further guidance on how to tailor V-HAARF for different types of processes within Volvo Cars.

A.3.1 General Adaptation Principles

When adapting V-HAARF to a specific domain, the following principles should be considered:

- Risk and Compliance Level: Domains with higher regulatory or safety requirements should place greater emphasis on Layer 1 (Strategic Evolution) and governance aspects.
- Process Complexity: More complex processes may require deeper analysis in Layer 2 (Process Readiness), especially regarding process debt.
- Task Nature: Processes with high interpretive or reasoning requirements benefit from stronger focus on Layer 3 (Task Suitability)
- Strategic Importance: Processes that are critical to business goals or digital transformation should receive more rigorous evaluation across all layers.

A.3.2 Domain-Specific Adaptation Recommendations

The table below provides guidance on how to adapt the framework for different types of domains commonly found at Volvo Cars:

Domain Type	Example Areas	Recommended Layer Emphasis	Key Adaptation Points	Suggested Weight Adjustment
Regulated	Crash testing, Compliance, Functional Safety	**Layer 1 + Layer 2**	Strong focus on governance, auditability, compliance, and process debt	Increase Layer 1 weight to 40%
Engineering & Verification	Test planning, Virtual simulation, Diagnostics	**Layer 2 + Layer 3**	Detailed process mapping and task-level analysis (T-IPO + LARA)	Keep default or slightly increase Layer 3
Operational / Manufacturing	Production support, Quality control, Logistics	**Layer 2 + Layer 3**	Focus on process efficiency, data operations, and effort vs. benefit analysis	Increase Layer 3 weight to 40%
Administrative / Support	Procurement, Finance, HR processes	**Layer 1 + Layer 2**	Emphasis on strategic alignment and reducing process debt	Increase Layer 1 weight
Data-Intensive / Analytics	Predictive maintenance, Data platforms	**Layer 2 + Layer 3**	Strong focus on data quality, control flow, and task automation potential	Increase Layer 2 and Layer 3

A.3.3 How to Adapt Each Layer

Layer 1 – Strategic Evolution Layer
In high-risk or regulated domains, evaluate leadership commitment and alignment more strictly.

In operational domains, focus more on whether the strategic vision supports efficiency and scalability.

Layer 2 – Process Readiness Layer
In complex engineering processes, conduct deeper analysis of the five perspectives (especially Data Operations and Control Flow).

In administrative processes, pay special attention to identifying and reducing process debt.

Layer 3 – Task Suitability Layer
In domains with many repetitive or rule-based tasks, give more weight to automation suitability.

In domains requiring high interpretation or decision-making (e.g., diagnostics), focus more on agentic AI potential using T-IPO and LARA principles.

A.3.4 Practical Recommendations for Adaptation

- **Adjust Layer Weights**: Modify the default weights (30%-40%-30%) based on domain risk and strategic priority.
- **Customize Evaluation Criteria**: Add or emphasize domain-specific criteria (e.g., safety standards, regulatory compliance, or data sensitivity).
- **Involve Domain Experts**: Always include subject matter experts from the relevant domain during workshops and scoring.
- **Document Adaptations**: Clearly record any changes made to weights or criteria for transparency and future reference.
- **Start Simple**: For the first application in a new domain, begin with default settings and adjust in subsequent iterations.

B. Appendix

B.1 Scoring Criteria for the Post-Implementation Evaluation Framework (1–5 Scale)

Score	Description	Interpretation
5	Excellent	High maturity, strong governance, and clear business value
4	Good	Strong performance with minor improvement areas
3	Moderate	Acceptable performance but requires noticeable improvements
2	Low	Significant weaknesses in one or more layers
1	Very Low	Major issues across multiple layers; high risk

Layer-wise Evaluation Focus:

- Layer 1 (Lifecycle & Governance): Auditability, compliance, knowledge transfer, and sustainability.
- Layer 2 (Developer & Maintainability): Learning cost, maintainability, development efficiency, and technical quality.
- Layer 3 (Real-World Performance & Impact): Accuracy, robustness, user experience, and business value delivered.

B.2 Issue Taxonomy and Checklist

The Issue Taxonomy provides a structured way to categorise problems commonly encountered during the evaluation of agentic AI systems. Issues are grouped under three main dimensions aligned with the post-implementation evaluation framework.

Dimension	Issue Category	Description	Typical Indicators	Priority Level
Governance & Lifecycle Issues	Auditability & Traceability	Lack of proper logging or traceability of AI decisions and actions	Missing or incomplete decision logs	High
Governance & Lifecycle Issues	Knowledge Transfer & Reuse	Failure to capture and store successful user corrections or optimisations	Optimisations not saved for future use	High
Governance & Lifecycle Issues	Compliance & Risk Management	Insufficient controls to meet internal governance or regulatory standards	Weak escalation mechanisms or risk documentation	High
Maintainability & Development Issues	Response Latency & Performance	Slow or inconsistent system response times	Long waiting times before receiving output	Medium
Maintainability & Development Issues	Edge Case & Exception Handling	Poor handling of unusual, rare, or unexpected inputs	System failures on non-standard inputs	Medium
Maintainability & Development Issues	Maintainability & Technical Debt	Difficulty in modifying, scaling, or maintaining the AI system over time	High effort required for small changes	Medium
Real-World Performance Issues	Accuracy & Consistency	Incorrect, inconsistent, or unreliable outputs	Frequent mismatches between expected and actual results	High
Real-World Performance Issues	Usability & User Experience	Confusing feedback, unclear responses, or poor user interaction design	Inconsistent messages or difficult recovery from errors	Medium
Real-World Performance Issues	Explainability & Feedback	Limited ability for users to understand or provide feedback on AI decisions	Black-box behaviour or limited feedback channels	Medium

B.3 Governance and Risk SLO Checklist

The Governance and Risk SLO Checklist helps evaluate whether an agentic AI system meets minimum governance, compliance, and risk management standards.

	Governance / Risk Area	Description / Requirement	Recommended Acceptance Criteria	Suggested Mitigation Action	Status
1	Auditability of Decisions	All AI actions and recommendations must be logged and traceable	Full logging of every operation	Implement structured back-end logging	-
2	Knowledge Transfer & Reuse	Successful corrections and optimisations should be stored for future reuse	> 80% of key issues/optimisations stored	Enable systematic back-end storage mechanism	-
3	Data Privacy & Compliance	No sensitive data should be exposed or processed without proper controls	Full compliance with internal data policies	Apply data masking, access control, and anonymisation	-
4	Response Latency	System should respond within acceptable time limits for interactive use	Average response time < 3 seconds	Optimise prompt engineering and model selection	-
5	Error Rate & Robustness	The system should maintain an acceptable failure rate, especially for critical operations	Critical error rate < 5%	Add fallback mechanisms and user guidance	-
6	Human Oversight & Escalation	Clear mechanisms should exist for human review and escalation when required	Defined escalation paths for high-risk cases	Implement human-in-the-loop checkpoints	-
7	Transparency & Explainability	Users should be able to understand the basis of AI recommendations	Key decisions should be explainable	Improve response clarity and add reasoning summaries	-
8	Overall Risk Level	Combined assessment of governance, technical, and operational risks	Low to Medium risk	Conduct formal risk review before scaling or production	-

C. Appendix

C.1 Summary of Empirical Data Collected

Number of issues logged: 42
Validation reports analyzed: 149 test cases
Evaluation period: January – June 2026

C.2 Scoring Summary – CTPBot Project

Layer	Score (1–5)	Weight	Weighted Score
Layer 1: Lifecycle & Governance	4.0	30%	1.20
Layer 2: Developer & Maintainability	3.2	40%	1.28
Layer 3: Real-World Performance	3.9	30%	1.17
Overall: AI System Effectiveness Score	-	-	**73 / 100**

C.3 Key Findings from the Evaluation

- Strengths identified in governance awareness and basic functionality.
- Improvement areas: Response latency, handling of edge cases, and certain maintainability aspects.
- The framework successfully identified both technical and organizational improvement opportunities.

D. Appendix

Templates and Tools

This appendix presents the practical tools and templates developed for the implementation of the Volvo Hybrid AI Adoption & Readiness Framework (V-HAARF) and the Post-Implementation Evaluation Framework. The purpose of the development of these tools was to increase consistency, reduce manual work and enable repeatable evaluations across different projects at Volvo Cars.

D.1 Evaluation Scoring Template Structure

An Excel-based Evaluation Scoring Template was developed to support the scoring process for both frameworks. The template automates calculations and provides visual summaries to help evaluators interpret results quickly.

Main Features of the Template:

- Layer-wise Scoring Sheets: Separate sheets for each layer (Strategic/Process/Task for V-HAARF, and Governance/Maintainability/Real-World Performance for the post-implementation framework).
- 1–5 Scoring Input: Users input scores from 1 to 5 along with supporting evidence or comments.
- Automatic Weighted Scoring: The template automatically calculates the overall score using the predefined weighting formula.
- Visual Outputs: Includes radar charts (especially for Layer 2 dimensions) and summary dashboards.
- Evidence Logging: Space to document observations, data sources, and justification for each score.
- Recommendation Section: A dedicated area to record key findings and suggested actions.

The template was designed to be flexible so it can be used for both pre-adoption evaluations (using V-HAARF) and post-implementation assessments.

D.2 Effort Estimation Model Template

An Effort and Complexity Estimation Model was created to help teams estimate the time and resources required when applying the frameworks. The model breaks down the evaluation and improvement process into clear phases and provides indicative effort ranges.

Structure of the Model:

The model is organised around the following main phases:

Phase	Description	Typical Effort Driver	Notes
Onboarding & Documentation Review	Understanding the process/AI system and collecting initial documentation	Project complexity	One-time per project
Scenario-based Testing / Evaluation	Running test scenarios or structured evaluation of the system	Number and complexity of use cases	Core effort phase
Issue Logging & Analysis	Documenting issues, categorising them, and performing root cause analysis	Volume and severity of issues	Highest effort phase
Validation & Refinement	Reviewing findings and refining scores or recommendations	Quality of available data	Can be reduced with better tooling
Framework Application & Reporting	Applying the framework, calculating scores, and preparing recommendations	Scope and depth of analysis	Includes stakeholder workshops
Knowledge Transfer & Documentation	Capturing lessons learned and storing improvements for future use	Organisational learning goals	Supports long-term value

Key Characteristics of the Model:

- Provides average effort ranges (in hours or person-days) based on typical industrial projects.
- Includes scaling factors that allow adjustment according to project size, complexity, and domain risk level.
- Helps teams create realistic plans and business cases when proposing AI-related evaluations or improvements.
- Can be used during both the initial application of V-HAARF and during post-implementation reviews.